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Substrate cleaning apparatus and substrate cleaning method

Abstract

A particle removed from a substrate is suppressed from adhering to the substrate again. A substrate cleaning apparatus includes a substrate holder configured to hold the substrate; a gas nozzle configured to jet a cleaning gas to the substrate on the substrate holder; and a nozzle cover provided to surround the gas nozzle. The cleaning gas is jetted to a decompression chamber of the nozzle cover from the gas nozzle, and a gas cluster configured to remove the particle on the substrate in the decompression chamber is generated. A gas for a gas curtain is jetted from a holder support of the substrate holder toward the nozzle cover, and the gas curtain is formed between the nozzle cover and the holder support.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This Application is a U.S. national phase application under 35 U.S.C. § 371 of PCT Application No. PCT/JP2020/039244 filed on Oct. 19, 2020, which claims the benefit of Japanese Patent Application No. 2019-200157 filed on Nov. 1, 2019, the entire disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The various aspects and embodiments described herein pertain generally to a substrate cleaning apparatus and a substrate cleaning method.

BACKGROUND

(3) In a semiconductor manufacturing apparatus, adhesion of particles to a substrate during a manufacturing process is one of major factors that decide the yield of products. For this reason, the substrate is cleaned before or after it is subjected to a processing. In this regard, it is required to develop a cleaning technique enabling to remove particles reliably by a simple method while suppressing damage to the substrate. Various cleaning techniques have been researched and developed to peel the particles from a surface of the substrate by applying a physical shearing force equal to or larger than the adhesive strength between the particles and the substrate. One of these techniques is a technique using a physical shearing force of a gas cluster.

(4) The gas cluster is an aggregate (cluster) in which a large number of atoms or molecules are aggregated when a high-pressure gas is jetted into a vacuum formed in a decompression chamber or the like to be cooled to a condensation temperature by adiabatic expansion. When cleaning the substrate, the particles are removed by radiating the gas cluster to the substrate while appropriately accelerating it or without accelerating it.

(5) Further, conventionally, a technique for effectively removing particles attached in a pattern on the surface of the substrate has been developed (see Patent Document 1).

(6) In this case, if the particles removed from the substrate can be suppressed from being wound up and re-adhering to the substrate, the substrate can be cleaned more effectively. Patent Document 1:

SUMMARY

(7) In an exemplary embodiment, a substrate cleaning apparatus includes a substrate holder configured to hold a substrate; a nozzle cover equipped with a decompression chamber configured to form a decompressed atmosphere with respect to the substrate; and a gas nozzle configured to jet a cleaning gas having a pressure higher than that of the decompression chamber to generate, in the decompression chamber, a gas cluster configured to clean the substrate. The nozzle cover comprises a nozzle cover main body equipped with the decompression chamber and a flange provided at a lower end periphery of the nozzle cover main body. The substrate holder is provided with a gas curtain forming unit configured to form a gas curtain by jetting a gas for the gas curtain to the flange.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a side view illustrating a substrate cleaning apparatus according to an exemplary embodiment.
- (2) FIG. 2 is a side view illustrating a gas nozzle and a nozzle cover of the substrate cleaning apparatus shown in FIG. 1.
- (3) FIG. 3 is a side view illustrating an operation of the substrate cleaning apparatus.
- (4) FIG. 4A to FIG. 4D are side views illustrating a state in which a particle is removed by a gas cluster.
- (5) FIG. 5A to FIG. 5D are side views illustrating a state in which a particle is removed by a gas cluster.
- (6) FIG. 6 is a plan view illustrating a surface of a wafer.
- (7) FIG. 7 is a diagram illustrating an entire vacuum processing apparatus in which the substrate cleaning apparatus is provided.
- (8) FIG. 8 is a diagram illustrating a modification example of a gas flow path, and is similar to FIG. 2.
- (9) FIG. 9 is a diagram illustrating a modification example of a substrate holder.

DETAILED DESCRIPTION

Exemplary Embodiment

(10) First, an entire vacuum processing apparatus equipped with a substrate cleaning apparatus according to an exemplary embodiment will be described with reference to FIG. 7. FIG. 7 is a plan view showing an overall configuration of a vacuum processing apparatus **101** which is a multi-chamber system. In the vacuum processing apparatus **101**, carry-in/out ports **112** each configured to place thereon a FOUP **111** as a sealed transfer container in which, for example, twenty five sheets of semiconductor wafers (hereinafter referred to as “wafers”) **W** as substrates are accommodated are arranged side by side in a transversal direction at, for example, three locations. Further, an atmospheric transfer chamber **113** is provided along the arrangement of these carry-in/out ports **112**, and a gate door **GT** configured to be opened or closed along with a cover of the FOUP **111** is provided on a front wall of the atmospheric transfer chamber **113**.

(11) Two load lock chambers **114** and **115**, for example, are airtightly connected to the atmospheric transfer chamber **113** on the opposite side to the carry-in/out ports **112**. A vacuum pump (not shown) and a leak valve (not shown) are provided in each of these load lock chambers **114** and **115** to perform a switchover between a normal pressure atmosphere and a vacuum atmosphere. Further, in FIG. 10, gate valves (sluice valves) **G** are provided. Moreover, the atmospheric transfer chamber **113** is provided with a first substrate transfer device **116** composed of a multi-joint arm for transferring the wafer **W**. Further, when viewed from the front side to the rear side of the

atmospheric transfer chamber **113**, a wafer inspection unit **117** serving as a substrate inspection unit is provided on a right wall of the atmospheric transfer chamber **113**, and an alignment chamber **118** configured to adjust direction and eccentricity of the wafer W is provided on a left wall of the atmospheric transfer chamber **113**. The first substrate transfer device **116** serves to transfer the wafer W to/from the FOUP **111**, the load lock chambers **114** and **115**, the wafer inspection unit **117**, and the alignment chamber **118**. For this reason, the first substrate transfer device **116** is configured to be movable in, for example, the arrangement direction (X direction in FIG. 1) of the FOUPs **111**, movable up and down, rotatable about a vertical axis, and movable back and forth.

(12) When viewed from the atmospheric transfer chamber **113**, a vacuum transfer chamber **102** is hermetically connected to the back side of the load lock chambers **114** and **115**. This vacuum transfer chamber **102** is connected with five vacuum processing modules **121** to **125** and a substrate cleaning apparatus **100** according to the present exemplary embodiment. The vacuum processing modules **121** to **125** are configured as vacuum processing modules configured to perform a sputtering processing and a CVD (Chemical Vapor Deposition) processing for film formation including copper wiring on the wafer W having, as recesses for a circuit pattern formation, via holes and grooves for burying the copper wiring.

(13) Further, the vacuum transfer chamber **102** is equipped with a second substrate transfer device **126** configured to transfer the wafer W in the vacuum atmosphere, and the wafer W is transferred to/from the load lock chambers **114** and **115**, a substrate cleaning apparatus **100**, and the vacuum processing modules **121** to **125** by the second substrate transfer device **126**. The second substrate transfer device **126** is equipped with a multi-joint arm **126a** configured to be rotatable about a vertical axis and movable back and forth, and this multi-joint arm **126a** is configured to be moved in a lengthwise direction (Y direction in FIG. 1) by a base **126b**.

(14) Next, the wafer inspection unit **117** will be explained. The wafer inspection unit **117** is configured to acquire particle information including a particle diameter for a particle adhering to the wafer W. The particle information is information for detecting, for example, a position and a size of the particle on the wafer W. An apparatus capable of evaluating the particle diameter of the particle on the surface of the wafer, for example, an optical or electron beam type surface defect inspection apparatus using specular reflection light or scattered light may be used as the wafer inspection unit **117**. Alternatively, a scanning probe microscope such as a scanning electron microscope (SEM), a scanning tunnel microscope (STM), or an atomic force microscope (AFM) may be used.

(15) Now, the substrate cleaning apparatus **100** according to the present exemplary embodiment will be described. The substrate cleaning apparatus **100** is configured to remove a deposit adhering to the wafer W accommodated therein.

(16) As depicted in FIG. 1 to FIG. 3, the substrate cleaning apparatus **100** includes: a cleaning chamber **31**; a substrate holder **11** disposed in the cleaning chamber **31** and configured to hold the wafer W, which is placed thereon horizontally, in a rotatably manner; a gas nozzle **50** configured to jet a cleaning gas to the wafer W on the substrate holder **11**; and a nozzle cover **20** surrounding the gas nozzle **50** and equipped with a decompression chamber **20A** in which a decompressed atmosphere is formed with respect to the wafer W.

(17) Among these, the cleaning chamber **31** is formed to have a positive pressure with respect to the exterior air. A clean gas supply **40** configured to supply a clean gas is provided at an upper portion of the cleaning chamber **31**, and this clean gas supply **40** is connected to a gas source **42** via a gas supply line **41**.

(18) Further, a gas exhaust device **44** such as an exhaust fan is connected to a lower portion of the cleaning chamber **31** via a gas exhaust line **43**. In addition, an opening **34** for carry-in or carry-out of the wafer W is provided at a sidewall of the cleaning chamber **31**, and the opening **34** is hermetically sealed so as to be opened or closed by a gate **35**.

(19) Further, the substrate holder **11** is equipped with a holder main body **11A** configured to hold

the wafer W placed thereon horizontally, and a holder support **12** disposed around the holder main body **11A** to surround the wafer W from a lateral side thereof. The holder main body **11A** and the holder support **12** can be rotated as one body via a driving shaft **13** which is driven by a driving motor **14**.

(20) Further, the gas nozzle **50** is configured to jet a carbon dioxide (CO.sub.2) gas and a helium (He) gas to the wafer W as will be described later, and the nozzle cover **20** is provided so as to surround this gas nozzle **50**.

(21) Specifically, as shown in FIG. **1** to FIG. **3**, the nozzle cover **20** has an inverted cup shape, and includes a nozzle cover main body **21** having the decompression chamber **20A** therein and a flange **22** extending from a lower end periphery of the nozzle cover main body **21**. The gas nozzle **50** is mounted to this cup-shaped nozzle cover main body **21** so that it passes through a central portion of the nozzle cover main body **21**.

(22) The nozzle cover main body **21** and the flange **22** of the nozzle cover **20** are formed as one body, and are made of ceramic as a whole.

(23) Further, the nozzle cover **20** composed of the nozzle cover main body **21** and the flange **22** is provided so as to cover the entire region of the wafer W.

(24) Particularly, as will be described later, the nozzle cover **20** is configured to be movable with respect to the wafer W by a moving arm **17**. In the meanwhile, the decompression chamber **20A** of the nozzle cover **20** is moved from a center of the wafer W to a periphery thereof. In this case, while the decompression chamber **20A** is being moved from the center of the wafer W to the periphery thereof, the nozzle cover **20** can cover the entire region of the wafer W all the time. Accordingly, a particle removed from the wafer W is suppressed from being scattered to the outside of the nozzle cover **20**.

(25) Further, in the substrate holder **11**, an end portion **12A** of the holder support **12** on the nozzle cover **20** side is provided with jetting holes through which a gas for gas curtain such as a N.sub.2 gas or air supplied from a gas flow path **15** to be described later is jetted toward the nozzle cover **20**. For example, by allowing the holder support **12** of the substrate holder **11** to be made of a porous material, the end portion **12A** of the holder support **12** on the nozzle cover **20** side is provided with a large number of holes, and these holes are used as the jetting holes through which the gas for gas curtain is jetted toward the nozzle cover **20**. Thus, the holes at the end portion **12A** of the holder support **12** on the nozzle cover **20** side serve as a gas curtain forming unit configured to form a gas curtain between the holder support **12** and the nozzle cover **20**.

(26) Further, the gas for gas curtain composed of the N.sub.2 gas or the air is supplied to the holder support **12** through the gas flow path **15** connected to a gas source **16** for gas curtain. The gas flow path **15** may be directly provided from the driving shaft **13** to the holder support **12**, as shown in FIG. **2**, or may be configured to pass through the inside of the substrate holder **11** and the driving shaft **13** formed to be hollow, as illustrated in FIG. **8**.

(27) Further, in the nozzle cover main body **21** of the nozzle cover **20**, a decompression unit **24** configured to form the decompressed atmosphere in the decompression chamber **20A** of the nozzle cover **20** is provided adjacent to the gas nozzle **50**, and this decompression unit **24** is connected to a decompression pump **27** via a connection line **28**. In this case, the connection line **28** is extended through a wall of the cleaning chamber **31** and led to the decompression pump **27** provided outside the cleaning chamber **31**.

(28) Further, the gas nozzle **50** is held by the nozzle cover main body **21** of the nozzle cover **20**, and this gas nozzle **50** is horizontally moved in the cleaning chamber **31** along with the nozzle cover **20** by a moving arm **17** which is disposed in the cleaning chamber **31**. In the present exemplary embodiment, the gas nozzle **50** and the nozzle cover **20** are configured to be movable, above the wafer W held by the substrate holder **11**, from a center of the wafer W toward a periphery thereof or from the periphery of the wafer W toward the center thereof by the moving arm **17**.

(29) In addition, an outlet **50a** side of the gas nozzle **50** is enlarged in diameter. An outer diameter

L1 of a leading end of the outlet **50a** is set to be, e.g., 10 mm, and an outer diameter L2 of the end portion of the decompression chamber **20A** of the nozzle cover **20** on the wafer W side is set to be, e.g., 50 mm to 60 mm.

(30) As shown in FIG. 1 to FIG. 3, the substrate holder **11** configured to hold the wafer W is equipped with the holder main body **11A**, and the holder support **12** disposed around the holder main body **11A** to surround the wafer W from the lateral side thereof. Here, the holder main body **11A** is made of the SUS material, aluminum material or ceramic material, and has a high surface friction coefficient. Accordingly, just by placing the wafer W on the holder main body **11A**, the wafer W can be stably held on the holder main body **11A** without needing to provide an attraction mechanism.

(31) Moreover, it is desirable that the holder support **12** is made of a porous material such as ceramic and a top surface of the holder support **12** is on a level with a top surface of the holder main body **11A**. In this case, the porous material such as the ceramic may be exposed only in a portion of the holder support **12** facing the flange **22** of the nozzle cover **20**, whereas the other portions of the holder main body **11A** may be coated or a non-porous material may be stuck thereto. In this case, the gas for gas curtain may be supplied from the portion of the holder support **12** facing the flange **22**. Still alternatively, the holder support **12** may be made of a SUS material or an aluminum material instead of the ceramic material. The holder support **12** may be formed by machining the ceramic material, the SUS material, or the aluminum material.

(32) Further, as shown in FIG. 1, one end of a connection line **50b** extending through the wall of the cleaning chamber **31** is connected to the gas nozzle **50**. In the cleaning chamber **31**, the connection line **50b** has a flexible structure so it is capable of coping up with a movement of the gas nozzle **50**. In addition, the connection line **50b** is connected to a supply path **52** for a carbon dioxide (CO₂) gas and a supply path **53** for a helium (He) gas via a pressure control valve **51** which constitutes a pressure controller at the outside of the cleaning chamber **31**. The supply path **52** has an opening/closing valve V1, a carbon dioxide gas flow rate controller **52a**, and a carbon dioxide gas source **52b**, and the supply path **53** has an opening/closing valve V2, a helium gas flow rate controller **53a**, and a helium gas source **53b**.

(33) The carbon dioxide gas is a gas for cleaning (cleaning gas), and a gas cluster is formed by this gas. Further, the helium gas is a gas for pushing out (extruding gas). The helium gas is difficult to form the cluster, and if the helium gas is mixed into the carbon dioxide gas, the speed of the cluster generated by the carbon dioxide gas can be increased. Further, the connection line **50b** is provided with a pressure detector **54** configured to detect a pressure within the connection line **50b**. Based on a detection value of this pressure detector **54**, the opening degree of the pressure control valve **51** is adjusted by a controller **55** to be described later, so that a gas pressure within the decompression chamber **20A** is controlled.

(34) In addition, based on the detection value of the pressure detector **54**, the controller **55** may control the carbon dioxide gas flow rate controller **52a** and the helium gas flow rate controller **53a** to adjust flow rates of the gases. Further, a supply pressure of each gas may be increased by using a pressure raising mechanism such as a gas booster between the opening/closing valves V1 and V2 for the gases and the control valve **51**, and may be adjusted by the pressure control valve **51**.

(35) Moreover, as shown in FIG. 1 and FIG. 3, the vacuum processing apparatus **101** is provided with the controller **55** composed of, for example, a computer for controlling an overall operation of the whole apparatus. The controller **55** includes a CPU, a program, and a storage. The program has process groups created to implement an operation of the apparatus corresponding to a vacuum processing performed in the vacuum processing modules **121** to **125** in addition to a cleaning processing to be described later. The program is stored in a recording medium such as, but not limited to, a hard disk, a compact disk, a magneto-optical disk, a memory card or a flexible disk, and is installed from this recording medium to the controller **55**.

(36) Further, the particle information acquired by the wafer inspection unit **117** is stored in the

storage of the controller **55**. This particle information is information in which the position of the wafer **W** and the size of the particle are matched. The size of the particle is, for example, a value assigned according to the range of particle diameters set in the wafer inspection unit **117**, and the size is determined by values such as equal to or larger than 20 nm and less than 40 nm, or equal to or larger than 40 nm and less than 60 nm.

(37) Now, an operation of the present exemplary embodiment having the above-described configuration will be discussed.

(38) If the FOUP **111** is disposed in the carry-in/out port **112** shown in FIG. 7, the wafer **W** is taken out from the FOUP **111** by the first substrate transfer device **116**. This wafer **W** has, as pattern recesses, recesses (grooves and via holes) for burying a copper wiring. Then, the wafer **W** is transferred to the alignment chamber **118** via the atmospheric transfer chamber **113** in the normal pressure atmosphere to be aligned. Thereafter, the wafer **W** is transferred to the wafer inspection unit **117** by the first substrate transfer device **116**, and the particle information is acquired therein. The acquired particle information is sent to the controller **55**.

(39) The wafer **W** after being subjected to the inspection in the wafer inspection unit **117** is carried by the first substrate transfer device **116** into the load lock chamber **114** (**115**) set to be in the normal pressure atmosphere. Then, after the atmosphere within the load lock chamber **114** (**115**) is switched to the vacuum atmosphere, the wafer **W** is transferred to the substrate cleaning apparatus **100** by the second substrate transfer device **126** to be subjected to a particle removal processing.

(40) As depicted in FIG. 1 to FIG. 3, in the substrate cleaning apparatus **100**, the gas for gas curtain such as the N.sub.2 gas or the air supplied through the gas flow path **15** is first jetted to the nozzle cover **20** from the end portion of the holder support **12** of the substrate holder **11** on the nozzle cover **20** side (see FIG. 1). The gas for gas curtain jetted to the nozzle cover **20** forms a gas curtain **20B** between the holder support **12** and the flange **22** of the nozzle cover **20**, and seals between the inside of the decompression chamber **20A** of the nozzle cover **20** and the outside.

(41) In the meantime, the decompression pump **27** is operated, and the decompression chamber **20A** of the nozzle cover **20** is decompressed through the decompression unit **24**, so that the pressure within the decompression chamber **20A** is reduced as compared to a pressure at the outside of the nozzle cover **20**.

(42) Thereafter, the carbon dioxide gas is supplied from the gas nozzle **50** into the decompression chamber **20A** of the nozzle cover **20** as the cleaning gas, and the helium gas is supplied into the decompression chamber **20A** of the nozzle cover **20** as the extruding gas.

(43) In this case, by jetting the carbon dioxide gas as the cleaning gas from the outlet **50a** of the gas nozzle **50** into the decompression chamber **20A**, a gas cluster can be created in the decompression chamber **20A**, and a particle **1** adhering on the wafer **W** can be removed by using this gas cluster (see FIG. 6).

(44) In the present exemplary embodiment, an ejection amount **M1** of the gas for gas curtain jetted from the end portion of the holder support **12** on the nozzle cover **20** side is set to be smaller than an exhaust amount **M2** of the gas for gas curtain exhausted from the decompression chamber **20A** through the decompression unit **24**. By way of example, the ejection amount is set to be **M1**=10 L/min to 30 L/min, and the exhaust amount is set to be **M2**=**M1**×1 to 2 (1 to 2 times the ejection amount).

(45) Here, the principle of removing the particle from the surface of the wafer **W** using the gas cluster will be described. The gas cluster refers to a substance generated as atoms or molecules of a gas are gathered as an aggregate when the gas is supplied into a processing gas atmosphere from a region having a pressure higher than that of the decompression chamber **20A** of the nozzle cover **20** and cooled to a condensation temperature by adiabatic expansion. For example, a processing pressure in the decompression chamber **20A** of the nozzle cover **20** is set to be a vacuum atmosphere of, e.g., 0.1 Pa to 100 Pa, and the cleaning gas (carbon dioxide gas) is supplied at a pressure of 0.3 MPa to 5.0 MPa to the gas nozzle **50**. When this cleaning gas is supplied into the

processing atmosphere of the decompression chamber **20A** of the nozzle cover **20**, it is cooled to equal to or less than the condensation temperature due to rapid adiabatic expansion. As a result, molecules **2a** are combined with each other by a van der Waals force on the outlet **50a** side of the gas nozzle **50** to form a gas cluster **2** as the aggregate, as illustrated in FIG. 2 and FIG. 3. The gas cluster **2** is neutral in this example. By way of example, it is desirable that the gas cluster meets the condition of 5×10^{23} atoms (molecules)/cluster or more as it is about 8 nm in terms of 5×10^{23} atoms (molecules)/cluster.

(46) The gas cluster **2** generated on the outlet **50a** side of the gas nozzle **50** is vertically radiated toward the wafer **W**. Then, the gas cluster **2** enters the recess for circuit pattern on the wafer **W** and removes the particle **1** in the recess by blowing it away.

(47) A state in which the particle **1** on the wafer **W** is removed by the gas cluster **2** is schematically illustrated in FIG. 4A to FIG. 4D and FIG. 5A to FIG. 5D. FIG. 4A to FIG. 4D are diagrams illustrating a case where the gas cluster **2** directly collides with the particle **1** on the wafer **W**. In this case, the gas cluster **2** is radiated perpendicularly to the surface of the wafer **W** and is highly likely to collide with the particle **1** from obliquely above it, as illustrated in FIG. 4A. If the gas cluster **2** collides with the particle **1** in the state that it is offset from the particle **1** (that is, in the state that a center of the gas cluster **2** and a center of the particle **1** are deviated when viewed from above) as shown in FIG. 4B, a force that moves the particle **1** in a transversal direction is applied to the particle **1** by an impact at the time of the collision of the gas cluster **2**, as shown in FIG. 4C. As a result, the particle **1** is separated from the surface of the wafer **W**, floats up, and is blown sideways or obliquely upwards, as shown in FIG. 4D.

(48) Further, it is also possible to remove the particle **1** by radiating the gas cluster **2** to the vicinity of the particle **1** as shown in FIG. 5A without making the gas cluster **2** directly collide with the particle **1**. If the gas cluster **2** collides with the wafer **W**, the constituent molecules of the gas cluster **2** is decomposed while being spread in a transversal direction (see FIG. 5B). At this time, since a high kinetic energy density region moves in the transversal direction (horizontal direction), the particle **1** is separated from the wafer **W** to be blown away (see FIG. 5C and FIG. 5D). In this way, the particle **1** escapes from the recess and is dispersed into the decompression chamber **20A** of the nozzle cover **20**, and is then removed via the decompression unit **24** to the outside through the decompression pump **27** provided outside the cleaning chamber **31**.

(49) While the gas cluster **2** is being generated by jetting the carbon dioxide gas and the helium gas from the gas nozzle **50** to the wafer **W** and removing the particle **1** on the wafer **W** by using the generated gas cluster, the substrate holder **11** is rotated to rotate the wafer **W**, and the gas nozzle **50** and the nozzle cover **20** are moved from above the center of the wafer **W** to above the periphery thereof by the moving arm **17**. Thus, the particle **1** on the wafer **W** can be effectively removed by the gas cluster **2** over the entire region of the wafer **W**.

(50) In the meantime, when the gas nozzle **50** is located between the center and the periphery of the wafer **W** (see FIG. 1 and FIG. 2), the gas for gas curtain jetted from the end portion of the holder support **12** on the nozzle cover **20** side forms the gas curtain **20B** toward the nozzle cover in a gap **G** (for example, 0.85 mm) between the holder support **12** and the nozzle cover **20**, and thus maintains the inside of the decompression chamber **20A** of the nozzle cover **20** in a sealed state.

(51) Subsequently, if the gas nozzle **50** reaches the periphery of the wafer **W** (see FIG. 3), some of the gas for gas curtain jetted from the end portion of the holder support **12** on the nozzle cover **20** side is jetted to the nozzle cover **20**, and the rest of the gas is introduced into the decompression chamber **20A** as well. Since, however, a flow rate of the gas for gas curtain introduced into the decompression chamber **20A** is small, it does not affect the decompression atmosphere of the decompression chamber **20A**.

(52) In this case, the gas for gas curtain jetted from the end portion of the holder support **12** on the nozzle cover **20** side forms the gas curtain **20B** in the gap **G** between the nozzle cover **20** and the end portion of the holder support **12** on the nozzle cover **20** side, thus keeping the inside of the

decompression chamber **20A** of the nozzle cover **20** in the sealed state.

(53) In the present exemplary embodiment, the holder support **12** is fixed to the holder main body **11A** by non-illustrated clamp screws, and the height position of the holder support **12** can be adjusted based on the thickness of the wafer **W**. In FIG. **3**, as the height position of the holder support **12** is adjusted, the top surface of the wafer **W** becomes on a level with the top surface of the holder main body **11A**. Thus, the height of the top surface of the wafer **W** coincides with the height of the top surface of the holder support **12**.

(54) Accordingly, the gap **G** between the nozzle cover **20** and the end portion of the holder support **12** on the nozzle cover **20** side and the gap between the wafer **W** and the nozzle cover **20** are made same.

(55) According to the present exemplary embodiment as described above, while rotating the wafer **W** by the substrate holder **11**, the gas nozzle **50** and the nozzle cover **20** are moved by the moving arm **17** from the center of the wafer **W** toward the periphery thereof. Thus, the particle **1** on the wafer **W** can be removed by using the gas cluster **2** in the entire region of the wafer **W**. Further, while keeping the gas nozzle **50** and the nozzle cover **20** on the periphery of the wafer **W**, the wafer **W** is rotated by the substrate holder **11**. Thus, the particle **1** on the periphery of the wafer **W** can be removed by using the gas cluster **2**.

(56) In addition, by supplying the carbon dioxide gas and the helium gas from the gas nozzle **50** into the decompression chamber **20A** of the nozzle cover **20**, the gas cluster **2** is generated. The particle **1** on the wafer **W** is removed by using the gas cluster **2**, and the removed particle **1** is sent from the decompression chamber **20A** to the outside. For this reason, the particle **1** removed from the wafer **W** is suppressed from being wound up to adhere to other portions of the wafer **W** again. Thus, the entire wafer **W** can be kept clean.

(57) Particularly, the nozzle cover **20** is configured to be movable with respect to the wafer **W** by the moving arm **17**. While removing the particle **1** on the wafer **W** by using the gas cluster **2**, the decompression chamber **20A** of the nozzle cover **20** is moved from the center of the wafer **W** to the periphery thereof. In this case, while the decompression chamber **20A** is being moved from the center of the wafer **W** to the periphery thereof, the nozzle cover **20** covers the entire region of the wafer **W** all the time, thus suppressing the particle **1** removed from the wafer **W** from being scattered to the outside of the nozzle cover **20**. Thus, the particle **1** can be securely suppressed from being wound up and adhering to the wafer **W** again.

(58) In addition, in the above-described exemplary embodiment, although the carbon dioxide gas is used as the cleaning gas and the helium gas is used as the extruding gas, the present exemplary embodiment is not limited thereto. By way of example, an argon gas or the like may be used as the cleaning gas, and a hydrogen gas or the like may be used as the extruding gas. When a combination of the carbon dioxide gas and the hydrogen gas is used, a high cleaning effect can be obtained with a relatively low-priced gas.

(59) Furthermore, although the above exemplary embodiment has been described for the example where the gas for gas curtain such as the N.sub.2 gas or the air is jetted to the wafer **W** from the end portion **12A** of the holder support **12** on the nozzle cover **20** side, the present exemplary embodiment is not limited thereto, and other gases may be used as long as they do not affect the operation of generating the gas cluster.

(60) Moreover, in the above-described exemplary embodiment, the gas nozzle **50** and the nozzle cover **20** are moved by the moving arm **17** above the wafer **W** held by the substrate holder **11**. However, other configurations may be adopted as long as the gas nozzle **50** and the nozzle cover **20** can be moved relatively above the wafer **W**. By way of example, as depicted in FIG. **9**, a driving mechanism **60** having a driving unit **60a** and a driving arm **60b** may be provided, and the substrate holder **11** may be connected to the driving arm **60b**. In this case, the substrate holder **11** is moved horizontally by the driving mechanism **60** while the wafer **W** held by the substrate holder **11** is being rotated as the substrate holder **11** is driven by the driving motor **14**. Therefore, the entire

surface of the wafer W can be cleaned without needing to move the gas nozzle 50 and the nozzle cover 20.

(61) According to the exemplary embodiment, it is possible to suppress the particles removed from the substrate from adhering to the substrate again.

Claims

1. A substrate cleaning apparatus, comprising: a substrate holder configured to hold a substrate; a nozzle cover equipped with a decompression chamber in which a decompressed atmosphere is formed with respect to the substrate; and a gas nozzle configured to jet a cleaning gas having a pressure higher than that of the decompression chamber to generate, in the decompression chamber, a gas cluster configured to clean the substrate, wherein the nozzle cover comprises a nozzle cover main body equipped with the decompression chamber and a flange provided at a lower end periphery of the nozzle cover main body, and the substrate holder is provided with a gas curtain forming unit configured to form a gas curtain by jetting a gas for the gas curtain to the flange, wherein the gas nozzle is configured to be horizontally moved relative to the substrate holder, and the nozzle cover covers an entire region of the substrate while the decompression chamber is being moved from a center of the substrate to a periphery thereof, and a decompression unit is formed in the nozzle cover and is connected to a decompression pump via a connection line, thereby setting an inside of the decompression chamber to be in the decompressed atmosphere.
2. The substrate cleaning apparatus of claim 1, wherein the substrate holder comprises a holder main body configured to hold the substrate, and a holder support disposed around the holder main body to surround a periphery of the substrate, and the gas curtain forming unit is provided at the holder support, and a gas flow path, which communicates with the gas curtain forming unit and is configured to supply the gas for the gas curtain to the gas curtain forming unit, is connected to the substrate holder.
3. The substrate cleaning apparatus of claim 1, wherein the substrate holder is configured to be rotated, and the gas nozzle and the substrate holder are relatively moved in a horizontal direction.
4. The substrate cleaning apparatus of claim 1, wherein the substrate holder comprises a holder main body configured to hold the substrate, and a holder support disposed around the holder main body to surround a periphery of the substrate, and the gas curtain forming unit is provided at the holder support, and a gas flow path, which communicates with the gas curtain forming unit and is configured to supply the gas for the gas curtain to the gas curtain forming unit, is connected to the substrate holder.
5. The substrate cleaning apparatus of claim 4, wherein the holder support is made of a porous material.
6. The substrate cleaning apparatus of claim 4, wherein a top surface of the substrate on the holder main body and a top surface of the holder support are on a level with each other.
7. The substrate cleaning apparatus of claim 1, wherein the substrate holder is configured to be rotated, and the gas nozzle and the substrate holder are relatively moved in a horizontal direction.
8. A substrate cleaning method performed in a substrate cleaning apparatus including a substrate holder configured to hold a substrate and a nozzle cover equipped with a decompression chamber in which a decompressed atmosphere is formed with respect to the substrate, the nozzle cover including a nozzle cover main body equipped with the decompression chamber and a flange provided at a lower end periphery of the nozzle cover main body, the substrate cleaning method comprising: forming a gas curtain by jetting a gas for the gas curtain to the flange from a gas curtain forming unit provided at the substrate holder; and generating a gas cluster configured to clean the substrate in the decompression chamber by jetting a cleaning gas having a pressure higher than that of the decompression chamber from a gas nozzle, wherein the gas nozzle is configured to be horizontally moved relative to the substrate holder, and the nozzle cover covers an entire region

of the substrate while the decompression chamber is being moved from a center of the substrate to a periphery thereof, and a decompression unit is formed in the nozzle cover and is connected to a decompression pump via a connection line, thereby setting an inside of the decompression chamber to be in the decompressed atmosphere.
